

Chapter 6: Trigonometry

Student Question Booklet

Year 10 Mathematical Methods : 2026

How to use this booklet

This booklet is your **map through Chapter 6**. For every section you will find:

- **Worked examples** — fully solved examples with a labelled diagram, plus the matching textbook example numbers and pages to cross-reference.
- **Question sets** — the questions to practise, grouped by *Building Understanding*, *Fluency*, *Problem-solving*, *Reasoning* and *Enrichment*, with page numbers. Find them in your own textbook.
- **Hurdle questions** — five questions you **must** be able to do to pass the section, each with a diagram and working space.

Diagrams are schematic (not to scale). Sections marked 10A are extension. Keep your calculator in **DEGREE** mode; round as instructed and always show working.

Page references are to *Essential Mathematics for the Victorian Curriculum 10 & 10A*, 3rd ed. Question sets belong to the textbook; the worked examples, diagrams and hurdle questions are original material written for this course.

6A Trigonometric ratios

CORE

Textbook worked examples (cross-reference in your textbook)

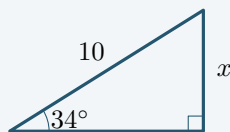
- **Example 1** — Solving for an unknown in the numerator (p. 496)
- **Example 2** — Solving for an unknown in the denominator (p. 497)

Question sets (practise from your textbook)

- **Building Understanding:** Q1–3 (p. 495)
- **Fluency:** Q1–3 (p. 498–500)
- **Problem-solving:** Q4–9 (p. 500–501)
- **Reasoning:** Q10–11 (p. 501)
- **Enrichment: Exploring identities:** Q12 (p. 501)

Worked examples (study the method, then try the hurdles)**Finding a side — unknown in the numerator**

Find x , correct to two decimal places. The hypotenuse is 10 cm, the marked angle is 34° , and x is the side *opposite* that angle.

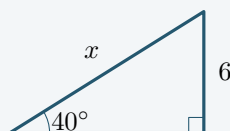


Solution. The unknown is opposite and the hypotenuse is known, so use **sine** ($\sin = \frac{\text{opp}}{\text{hyp}}$).

$$\sin 34^\circ = \frac{x}{10} \Rightarrow x = 10 \sin 34^\circ = 5.59 \text{ cm (2 dp)}.$$

Finding a side — unknown in the denominator

Find the hypotenuse x , correct to two decimal places, when the side *opposite* a 40° angle is 6 m.

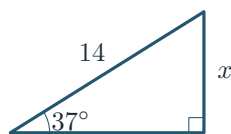


Solution. The unknown is the hypotenuse, so it sits in the denominator.

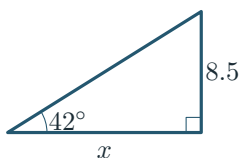
$$\sin 40^\circ = \frac{6}{x} \Rightarrow x = \frac{6}{\sin 40^\circ} = 9.33 \text{ m (2 dp)}.$$

Hurdle questions — Section 6A (show all working)

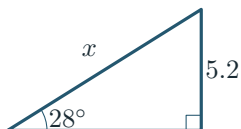
- H1.** Find the value of x , correct to two decimal places. A right-angled triangle has a hypotenuse of 14 cm and the angle between the hypotenuse and the base is 37° ; x is the side opposite the 37° angle.



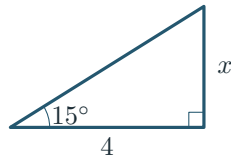
- H2.** Find the value of x , correct to two decimal places, where x is the *adjacent* side: the angle is 42° and the opposite side is 8.5 m. (*Hint: the unknown is in the denominator.*)



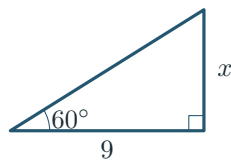
- H3.** A right-angled triangle has an angle of 28° and the side *opposite* this angle is 5.2 m. Find the length of the hypotenuse, correct to two decimal places.



- H4.** A skateboard ramp rises at an angle of 15° to the horizontal over a horizontal run of 4 m. Find the vertical height of the ramp, correct to two decimal places.



- H5.** Find the length of the unknown side marked x : the known angle is 60° , the side adjacent to it is 9 cm, and x is the side opposite the 60° angle. Give your answer correct to two decimal places.



6B Finding unknown angles

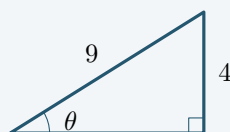
CORE

Textbook worked examples (cross-reference in your textbook)

- **Example 3** — Finding angles (p. 503)
- **Example 4** — Working with simple trigonometric applications (p. 504)

Question sets (practise from your textbook)

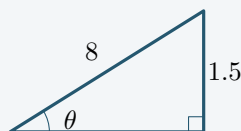
- **Building Understanding:** Q1–2 (p. 503)
- **Fluency:** Q1–3 (p. 505–506)
- **Problem-solving:** Q4–7 (p. 506)
- **Reasoning:** Q8–10 (p. 506–507)
- **Enrichment: A special triangle:** Q11 (p. 507)

Worked examples (study the method, then try the hurdles)**Finding an unknown angle**Find θ , correct to one decimal place, when the side opposite θ is 4 and the hypotenuse is 9.**Solution.** Opposite and hypotenuse are known, so use the inverse sine.

$$\sin \theta = \frac{4}{9} \Rightarrow \theta = \sin^{-1}\left(\frac{4}{9}\right) = 26.4^\circ \text{ (1 dp).}$$

Angle in an application

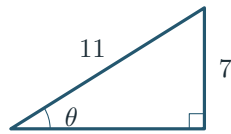
A ramp has a sloping surface 8 m long that reaches 1.5 m vertically. Find the angle the ramp makes with the horizontal, to one decimal place.

**Solution.** The 1.5 m rise is opposite the angle and the 8 m slope is the hypotenuse.

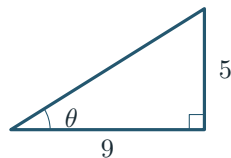
$$\sin \theta = \frac{1.5}{8} \Rightarrow \theta = \sin^{-1}(0.1875) = 10.8^\circ \text{ (1 dp).}$$

Hurdle questions — Section 6B (show all working)

- H1.** Find the angle θ , correct to one decimal place, in a right-angled triangle where the side opposite θ is 7 and the hypotenuse is 11.



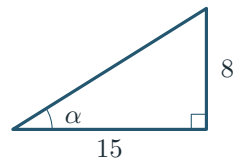
- H2.** Find the angle θ , correct to one decimal place, in a right-angled triangle where the side opposite θ is 5 and the adjacent side is 9.



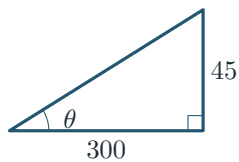
- H3.** A 6 m ladder leans against a wall, reaching 5.2 m up the wall. Find the angle the ladder makes with the ground, correct to one decimal place.



- H4.** A right-angled triangle has the two shorter sides (legs) of length 8 and 15. Find the sizes of *both* non-right angles, correct to one decimal place.



- H5.** A straight road climbs 45 m vertically over a horizontal distance of 300 m. Find the angle of incline of the road, correct to one decimal place.



6C Applications in two dimensions

CORE

Textbook worked examples (cross-reference in your textbook)

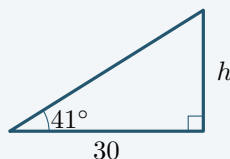
- **Example 5** — Working with an angle of elevation (p. 509)
- **Example 6** — Working with an angle of depression (p. 510)

Question sets (practise from your textbook)

- **Building Understanding:** Q1–2 (p. 509)
- **Fluency:** Q1–6 (p. 511)
- **Problem-solving:** Q7–9 (p. 511–512)
- **Reasoning:** Q10–13 (p. 512–513)
- **Enrichment: Vegetable garden design:** Q14 (p. 513)

Worked examples (study the method, then try the hurdles)**Angle of elevation**

From a point 30 m from the base of a tower on level ground, the angle of elevation to the top is 41° . Find the height h , to two decimal places.

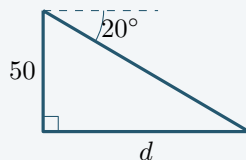


Solution. Height is opposite the angle and 30 m is adjacent, so use **tangent**.

$$\tan 41^\circ = \frac{h}{30} \Rightarrow h = 30 \tan 41^\circ = 26.08 \text{ m (2 dp)}.$$

Angle of depression

From the top of a 50 m cliff, the angle of depression to a boat is 20° . Find the horizontal distance d to the boat, to two decimal places.

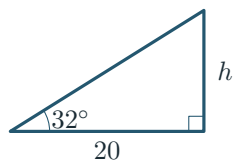


Solution. The depression angle equals the elevation angle from the boat. The 50 m height is opposite and d is adjacent.

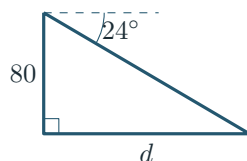
$$\tan 20^\circ = \frac{50}{d} \Rightarrow d = \frac{50}{\tan 20^\circ} = 137.37 \text{ m (2 dp)}.$$

Hurdle questions — Section 6C (show all working)

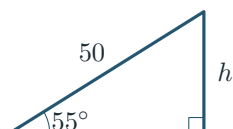
- H1.** From a point on level ground 20 m from the base of a tree, the angle of elevation to the top of the tree is 32° . Find the height of the tree, correct to two decimal places.



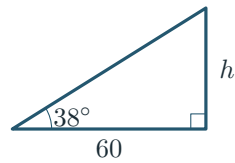
- H2.** From the top of a cliff 80 m high, the angle of depression to a boat at sea is 24° . Find the horizontal distance from the base of the cliff to the boat, correct to two decimal places.



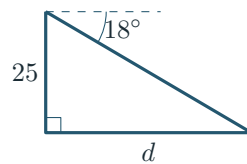
- H3.** A kite is flying on a taut string 50 m long that makes an angle of 55° with the horizontal. Find the height of the kite above the ground, correct to two decimal places.



- H4.** The angle of elevation to the top of a communications tower from a point 60 m from its base (on level ground) is 38° . Find the height of the tower, correct to two decimal places.



- H5.** From the top of a 25 m lighthouse, the angle of depression to a swimmer is 18° . Find the horizontal distance from the base of the lighthouse to the swimmer, correct to two decimal places.



6D Directions and bearings

CORE

Textbook worked examples (cross-reference in your textbook)

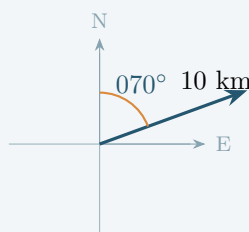
- **Example 7** — Stating a direction (p. 515)
- **Example 8** — Using bearings with trigonometry (p. 516)

Question sets (practise from your textbook)

- **Building Understanding:** Q1–3 (p. 515)
- **Fluency:** Q1–7 (p. 517–518)
- **Problem-solving:** Q8–11 (p. 518–519)
- **Reasoning:** Q12–14 (p. 519–520)
- **Enrichment: Navigation challenges:** Q15–16 (p. 520)

Worked examples (study the method, then try the hurdles)**Using a bearing with trigonometry**

A ship sails 10 km from port on a true bearing of 070° . Find how far *east* and how far *north* of the port it is, each to two decimal places.

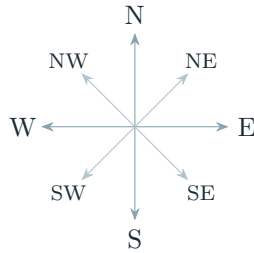


Solution. A bearing of 070° is 70° clockwise from north. Using the north–east right triangle:

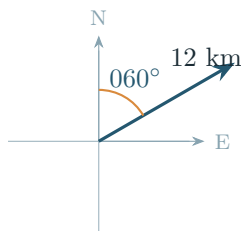
$$\begin{aligned}\text{east} &= 10 \sin 70^\circ = 9.40 \text{ km}, \\ \text{north} &= 10 \cos 70^\circ = 3.42 \text{ km}.\end{aligned}$$

Hurdle questions — Section 6D (show all working)

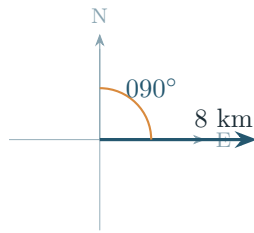
H1. Write the true bearing (three figures) of each direction: **(a)** due east, **(b)** $N40^\circ E$, **(c)** $S25^\circ W$.



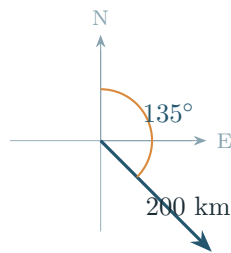
H2. A ship sails 12 km on a true bearing of 060° . Find how far *east* and how far *north* of its start it now is, each correct to two decimal places.



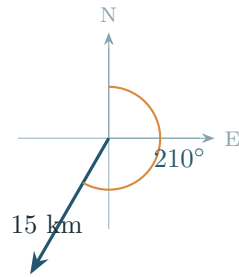
- H3.** Town B is 8 km due east of town A . State the true bearing of B from A , and the true bearing of A from B .



- H4.** A plane flies on a true bearing of 135° for 200 km. Find how far *south* and how far *east* it travels, each correct to one decimal place.



- H5.** From a port, a boat travels 15 km on a true bearing of 210° . Find how far *west* of the port the boat ends up, correct to two decimal places.



6E Applications in three dimensions

10A

Textbook worked examples (cross-reference in your textbook)

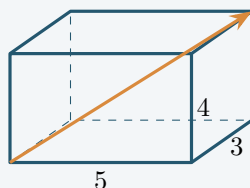
- **Example 9** — Applying trigonometry in 3D (p. 522)

Question sets (practise from your textbook)

- **Building Understanding:** Q1–2 (p. 522)
- **Fluency:** Q1–5 (p. 523–524)
- **Problem-solving:** Q6–9 (p. 524–525)
- **Reasoning:** Q10–11 (p. 524–525)
- **Enrichment: Three points in 3D:** Q12–13 (p. 526)

Worked examples (study the method, then try the hurdles)**Diagonal of a rectangular prism (3D)**

A rectangular prism measures $5 \times 3 \times 4$. Find the length of the space diagonal, and the angle it makes with the base.



Solution. First the base diagonal (Pythagoras on the 5×3 base):

$$d_{\text{base}} = \sqrt{5^2 + 3^2} = \sqrt{34} = 5.83.$$

The space diagonal rises 4 above this:

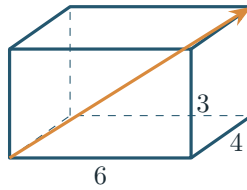
$$d_{\text{space}} = \sqrt{34 + 4^2} = \sqrt{50} = 7.07.$$

The angle α with the base uses height (opposite) and base diagonal (adjacent):

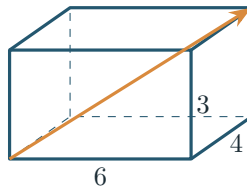
$$\tan \alpha = \frac{4}{5.83} \Rightarrow \alpha = 34.5^\circ \text{ (1 dp).}$$

Hurdle questions — Section 6E (show all working)

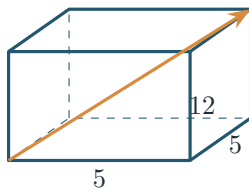
- H1.** A rectangular prism has dimensions $6\text{ cm} \times 4\text{ cm} \times 3\text{ cm}$. Find the length of the space diagonal (from one corner to the opposite corner), correct to two decimal places.



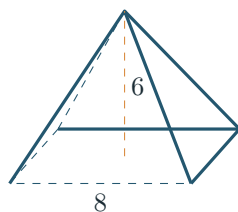
- H2.** For the prism in Question 1, find the angle that the space diagonal makes with the base (6×4 face), correct to one decimal place.



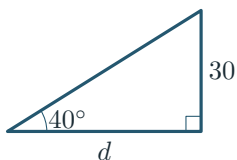
- H3.** A vertical flagpole 12 m tall stands at one corner of a flat rectangular courtyard measuring 5 m $\times$ 5 m. Find the angle of elevation of the top of the pole from the *opposite* corner of the courtyard, correct to one decimal place.



- H4.** A right pyramid has a square base of side 8 m and its apex is 6 m vertically above the centre of the base. Find the angle between a sloping edge and the base, correct to one decimal place.



- H5.** A tower stands vertically on level ground. From a point due south of the tower the angle of elevation to the top is 40° ; the top of the tower is 30 m high. Find the horizontal distance from this point to the base, correct to two decimal places.



6F The sine rule

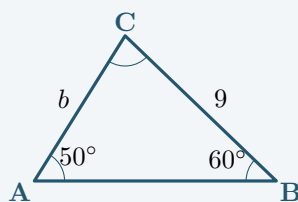
10A

Textbook worked examples (cross-reference in your textbook)

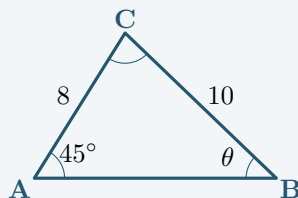
- **Example 10** — Finding a side length using the sine rule (p. 530)
- **Example 11** — Finding an angle using the sine rule (p. 530)

Question sets (practise from your textbook)

- **Building Understanding:** Q1–2 (p. 529)
- **Fluency:** Q1–2 (p. 531)
- **Problem-solving:** Q3–7 (p. 531–532)
- **Reasoning:** Q8–10 (p. 533)
- **Enrichment: More on the ambiguous case:** Q11 (p. 533)

Worked examples (study the method, then try the hurdles)**Finding a side with the sine rule**In triangle ABC , $A = 50^\circ$, $B = 60^\circ$ and $a = 9$. Find b , to one decimal place.**Solution.** Pair each side with its opposite angle: $\frac{b}{\sin B} = \frac{a}{\sin A}$.

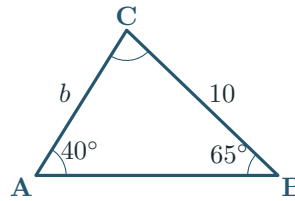
$$b = \frac{a \sin B}{\sin A} = \frac{9 \sin 60^\circ}{\sin 50^\circ} = 10.2 \text{ (1 dp).}$$

Finding an angle with the sine ruleIn triangle ABC , $a = 10$, $A = 45^\circ$ and $b = 8$. Find B , to one decimal place.**Solution.** Use $\frac{\sin B}{b} = \frac{\sin A}{a}$.

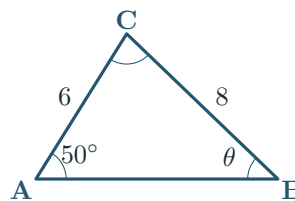
$$\sin B = \frac{b \sin A}{a} = \frac{8 \sin 45^\circ}{10} = 0.5657 \Rightarrow B = \sin^{-1}(0.5657) = 34.4^\circ.$$

Hurdle questions — Section 6F (show all working)

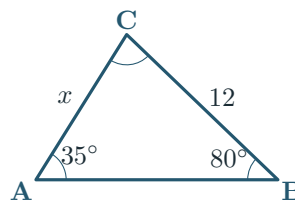
H1. In triangle ABC , $A = 40^\circ$, $B = 65^\circ$ and side $a = 10$. Find side b , correct to one decimal place.



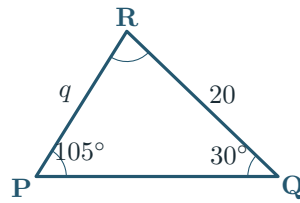
H2. In triangle ABC , $a = 8$, $b = 6$ and $A = 50^\circ$. Find angle B , correct to one decimal place.



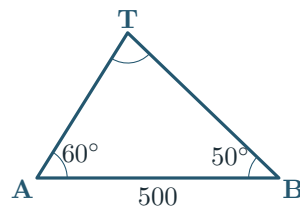
H3. A triangle has angles 35° and 80° , and the side of length 12 cm is opposite the 35° angle. Find the side opposite the 80° angle, correct to one decimal place.



- H4.** In triangle PQR , $P = 105^\circ$, $Q = 30^\circ$ and side $p = 20$ m. Find side q , correct to one decimal place.



- H5.** Two observers A and B are 500 m apart. From A the angle to a tree T is $\angle TAB = 60^\circ$; from B the angle is $\angle TBA = 50^\circ$. Find the distance AT , correct to one decimal place.



6G The cosine rule

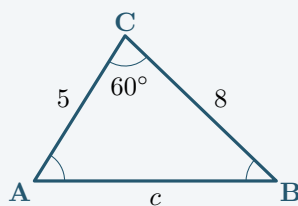
10A

Textbook worked examples (cross-reference in your textbook)

- **Example 12** — Finding a side length using the cosine rule (p. 535)
- **Example 13** — Finding an angle using the cosine rule (p. 536)

Question sets (practise from your textbook)

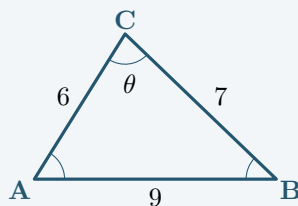
- **Building Understanding:** Q1–2 (p. 535)
- **Fluency:** Q1–2 (p. 536)
- **Problem-solving:** Q3–7 (p. 537)
- **Reasoning:** Q8–10 (p. 538)
- **Enrichment: Proof of the cosine rule:** Q11 (p. 538)

Worked examples (study the method, then try the hurdles)**Finding a side with the cosine rule**In triangle ABC , $a = 8$, $b = 5$ and the included angle $C = 60^\circ$. Find c exactly.**Solution.** Use $c^2 = a^2 + b^2 - 2ab \cos C$.

$$\begin{aligned}
 c^2 &= 8^2 + 5^2 - 2(8)(5) \cos 60^\circ \\
 &= 64 + 25 - 80(0.5) = 49, \\
 c &= 7.
 \end{aligned}$$

Finding an angle with the cosine rule

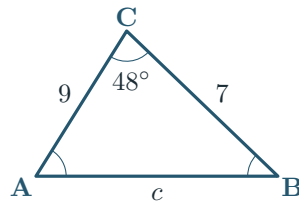
A triangle has sides 6, 7 and 9. Find the angle opposite the 9 side, to one decimal place.

**Solution.** Rearrange the cosine rule with $a = 6$, $b = 7$, $c = 9$.

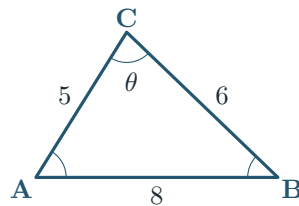
$$\cos C = \frac{6^2 + 7^2 - 9^2}{2(6)(7)} = \frac{4}{84} = 0.0476 \Rightarrow C = 87.3^\circ.$$

Hurdle questions — Section 6G (show all working)

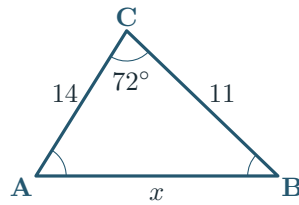
- H1.** In triangle ABC , $a = 7$, $b = 9$ and the included angle $C = 48^\circ$. Find side c , correct to two decimal places.



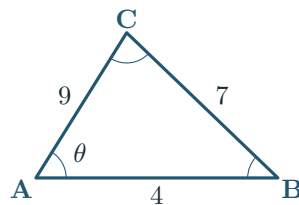
- H2.** A triangle has sides 5, 6 and 8. Find the size of the largest angle, correct to one decimal place. (*Which angle is the largest?*)



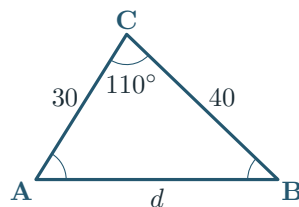
- H3.** Two sides of a triangle are 11 cm and 14 cm with an included angle of 72° . Find the length of the third side, correct to two decimal places.



- H4.** A triangle has sides 4 m, 7 m and 9 m. Find the angle opposite the 7 m side, correct to one decimal place.



- H5.** A ship sails 30 km, then turns so that the angle *inside* the triangle at the turn is 110° , and sails a further 40 km. Find its straight-line distance from the start, correct to two decimal places.



6H Area of a triangle

10A

Textbook worked examples (cross-reference in your textbook)

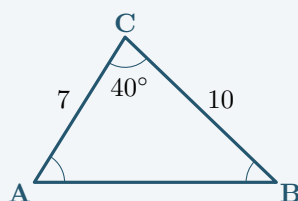
- **Example 14** — Finding the area of a triangle (p. 540)
- **Example 15** — Finding a side length given the area (p. 541)

Question sets (practise from your textbook)

- **Building Understanding:** Q1–2 (p. 540)
- **Fluency:** Q1–3 (p. 541)
- **Problem-solving:** Q4–7 (p. 542)
- **Reasoning:** Q8–10 (p. 543)
- **Enrichment: Polygon areas:** Q11 (p. 543)

Worked examples (study the method, then try the hurdles)**Area from two sides and the included angle**

Find the area of a triangle with sides 10 cm and 7 cm meeting at an angle of 40° , to two decimal places.

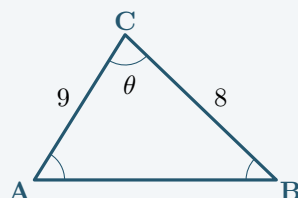


Solution. Use $\text{Area} = \frac{1}{2}ab \sin C$.

$$\text{Area} = \frac{1}{2}(10)(7) \sin 40^\circ = 22.50 \text{ cm}^2 \text{ (2 dp)}.$$

Finding the included angle from the area

A triangle has area 20 cm^2 with two sides 8 cm and 9 cm. Find the included angle, to one decimal place.

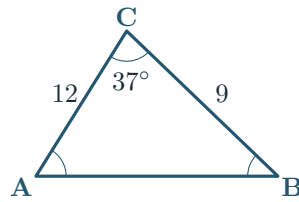


Solution. Substitute into $\text{Area} = \frac{1}{2}ab \sin C$ and solve for C .

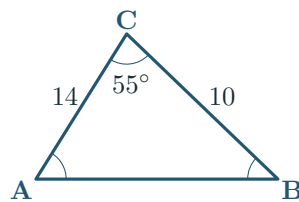
$$20 = \frac{1}{2}(8)(9) \sin C \Rightarrow \sin C = \frac{40}{72} = 0.5556 \Rightarrow C = 33.7^\circ.$$

Hurdle questions — Section 6H (show all working)

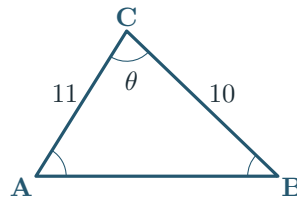
- H1.** Find the area of a triangle with two sides 9 cm and 12 cm and an included angle of 37° , correct to two decimal places.



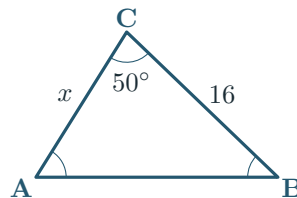
- H2.** Find the area of a triangle with two sides 10 m and 14 m and an included angle of 55° , correct to two decimal places.



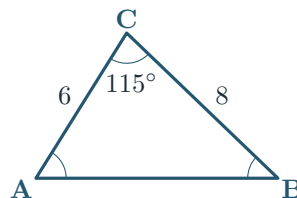
- H3.** A triangle has an area of 40 cm^2 . Two of its sides are 10 cm and 11 cm. Find the size of the included angle, correct to one decimal place.



- H4.** A triangle has area 60 m^2 , one side of 16 m, and the angle included between that side and an unknown side is 50° . Find the length of the unknown side, correct to two decimal places.



- H5.** A triangular garden bed has two sides of 8 m and 6 m meeting at an angle of 115° . Find the area of the garden bed, correct to two decimal places.



6I The unit circle

10A

Textbook worked examples (cross-reference in your textbook)

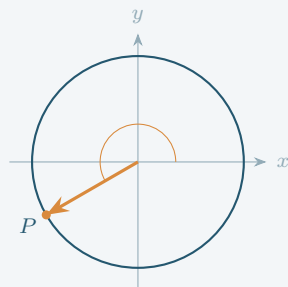
- **Example 16** — Choosing supplementary angles (p. 549)
- **Example 17** — Positioning a point on the unit circle (p. 550)
- **Example 18** — Using a reference angle (p. 551)

Question sets (practise from your textbook)

- **Building Understanding:** Q1–3 (p. 548)
- **Fluency:** Q1–5 (p. 552)
- **Problem-solving:** Q6–8 (p. 553)
- **Reasoning:** Q9–12 (p. 553–554)
- **Enrichment: Trigonometric identities:** Q13–14 (p. 554)

Worked examples (study the method, then try the hurdles)**Quadrant, signs and reference angle**

For $\theta = 210^\circ$: state the quadrant, the reference angle, and the sign of $\sin \theta$, $\cos \theta$ and $\tan \theta$. Hence write $\sin 210^\circ$ exactly.

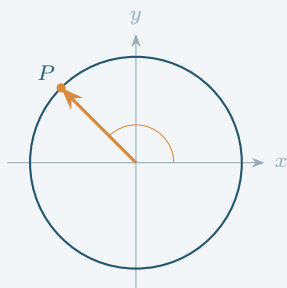


Solution. 210° lies in the **third quadrant**; its reference angle is $210^\circ - 180^\circ = 30^\circ$. In quadrant 3, $\sin$ and $\cos$ are negative and $\tan$ is positive. Therefore

$$\sin 210^\circ = -\sin 30^\circ = -\frac{1}{2}.$$

Using a reference angle for cosine

Write $\cos 135^\circ$ in terms of its reference angle and state its exact value.

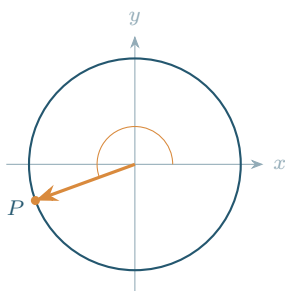


Solution. 135° is in the **second quadrant**; reference angle $180^\circ - 135^\circ = 45^\circ$. Cosine is negative in quadrant 2:

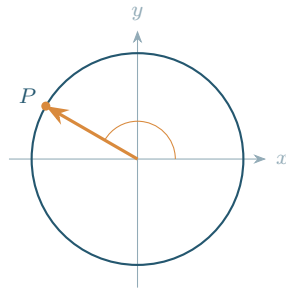
$$\cos 135^\circ = -\cos 45^\circ = -\frac{\sqrt{2}}{2} \approx -0.71.$$

Hurdle questions — Section 6I (show all working)

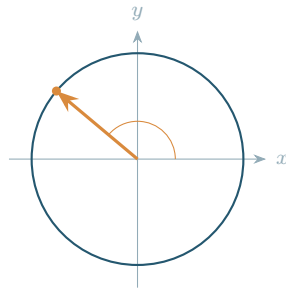
- H1.** For $\theta = 200^\circ$: state which quadrant θ lies in, and state whether $\sin \theta$, $\cos \theta$ and $\tan \theta$ are positive or negative.



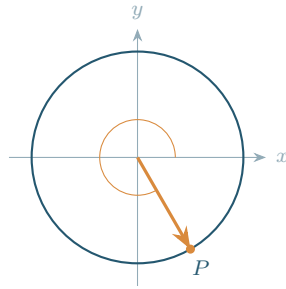
H2. Write $\sin 150^\circ$ in terms of the sine of its reference (acute) angle, then state its exact sign.



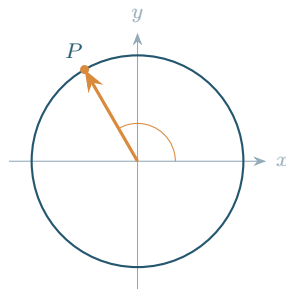
H3. Choose an obtuse angle θ that makes the statement true: $\sin \theta = \sin 40^\circ$.



H4. For $\theta = 300^\circ$: state the reference angle, and state whether $\cos 300^\circ$ is positive or negative.



H5. The point P on the unit circle corresponds to an angle of 120° . State the reference angle, and state the sign (positive or negative) of the x - and y -coordinates of P .



6J Graphs of trigonometric functions

10A

Textbook worked examples (cross-reference in your textbook)

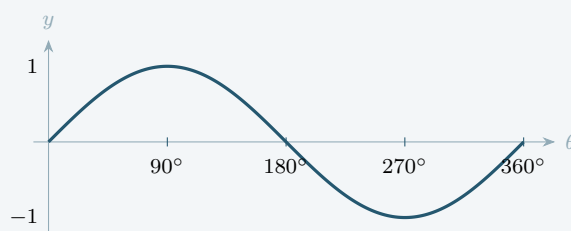
- **Example 19** — Reading off a trigonometric graph (p. 560)
- **Example 20** — Comparing the size of the sine of angles (p. 560)

Question sets (practise from your textbook)

- **Building Understanding:** Q1–2 (p. 559)
- **Fluency:** Q1–2 (p. 563)
- **Problem-solving:** Q3–6 (p. 564)
- **Reasoning:** Q7–9 (p. 564–565)
- **Enrichment: Trigonometric functions with technology:** Q10 (p. 565)

Worked examples (study the method, then try the hurdles)**Reading the sine graph**

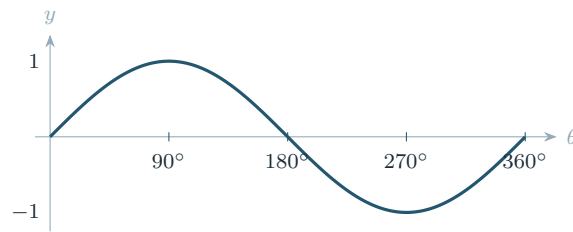
For $y = \sin \theta$ over $0^\circ \leq \theta \leq 360^\circ$, state the amplitude, the period, where the maximum occurs, and compare $\sin 100^\circ$ with $\sin 200^\circ$.



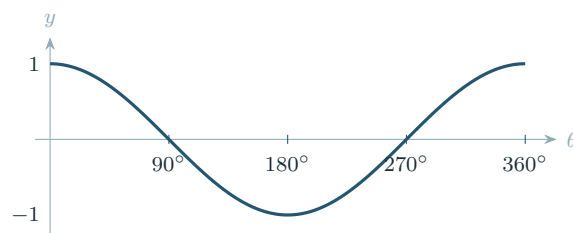
Solution. The curve rises to 1 and falls to -1 , so the amplitude is 1; one full cycle takes 360° , so the period is 360° . The maximum $y = 1$ occurs at $\theta = 90^\circ$. Since 100° is in quadrant 2 (sine positive) and 200° is in quadrant 3 (sine negative), $\sin 100^\circ > \sin 200^\circ$.

Hurdle questions — Section 6J (show all working)

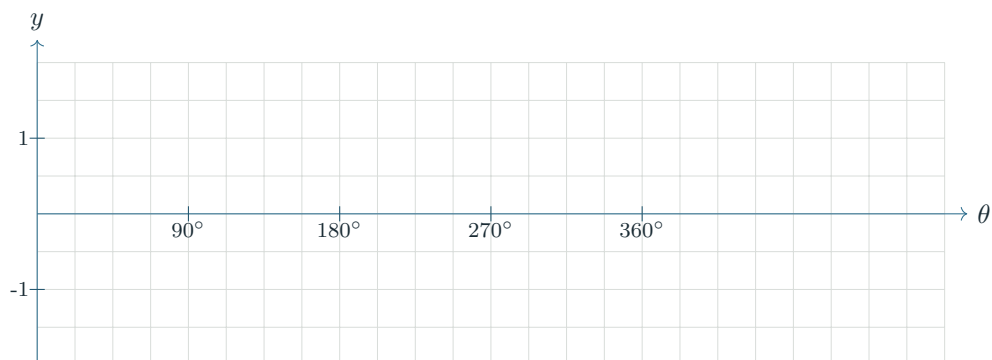
- H1.** For $y = \sin \theta$ over $0^\circ \leq \theta \leq 360^\circ$: state the maximum value, the minimum value, and *all* values of θ where the graph crosses the θ -axis.



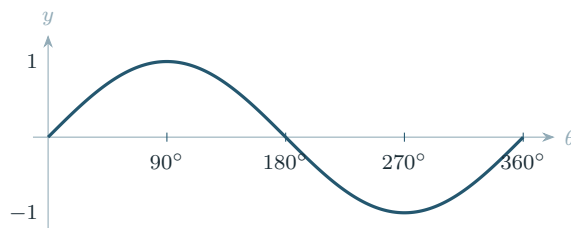
- H2.** State the amplitude and the period of the graph of $y = \cos \theta$.



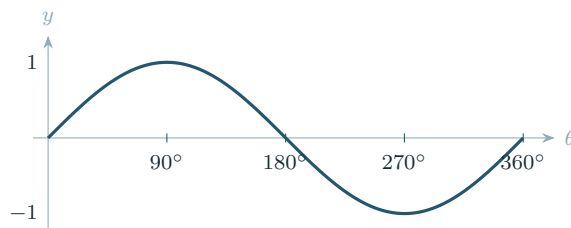
- H3.** On the grid provided, sketch $y = \cos \theta$ for $0^\circ \leq \theta \leq 360^\circ$. Clearly mark the θ -intercepts and the coordinates of the maximum and minimum points.



- H4.** Using your knowledge of the sine graph, estimate the value of $\sin 250^\circ$ to one decimal place, and state whether it is positive or negative.



- H5.** Without a calculator, state which is larger: $\sin 100^\circ$ or $\sin 200^\circ$. Justify your answer using the sign and shape of the sine graph.



6K Exact values and solving trigonometric equations

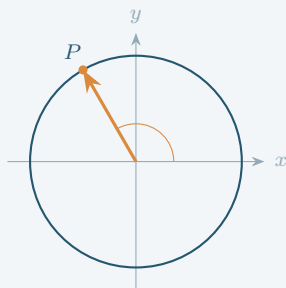
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Textbook worked examples (cross-reference in your textbook)

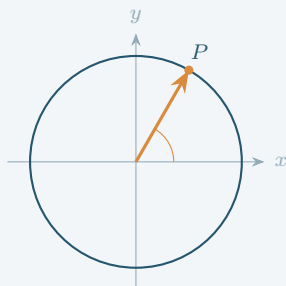
- **Example 21** — Using exact values (p. 568)
- **Example 22** — Solving simple trigonometric equations (p. 568)

Question sets (practise from your textbook)

- **Building Understanding:** Q1–3 (p. 567)
- **Fluency:** Q1–4 (p. 569)
- **Problem-solving:** Q5–7 (p. 570)
- **Reasoning:** Q8–9 (p. 570)
- **Enrichment: Rearranging trigonometric equations:** Q10–11 (p. 570)

Worked examples (study the method, then try the hurdles)**Exact value using a reference angle**Find the exact value of $\cos 120^\circ$.**Solution.** 120° is in quadrant 2 (reference angle 60°), where cosine is negative.

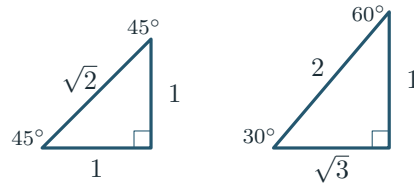
$$\cos 120^\circ = -\cos 60^\circ = -\frac{1}{2}.$$

Solving a trigonometric equationSolve $\cos \theta = \frac{1}{2}$ for $0^\circ \leq \theta \leq 360^\circ$.**Solution.** The reference angle is $\cos^{-1}(\frac{1}{2}) = 60^\circ$. Cosine is positive in quadrants 1 and 4, so

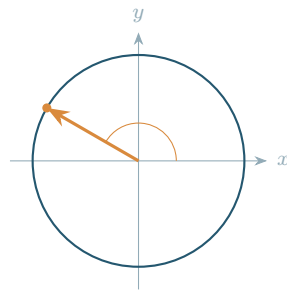
$$\theta = 60^\circ \quad \text{or} \quad \theta = 360^\circ - 60^\circ = 300^\circ.$$

Hurdle questions — Section 6K (show all working)

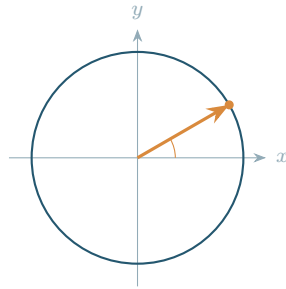
H1. Write down the *exact* values of: **(a)** $\sin 60^\circ$, **(b)** $\cos 30^\circ$, **(c)** $\tan 45^\circ$.



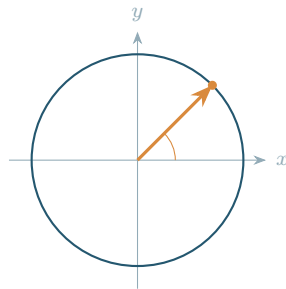
H2. Find the exact value of $\cos 150^\circ$, showing the reference angle you used.



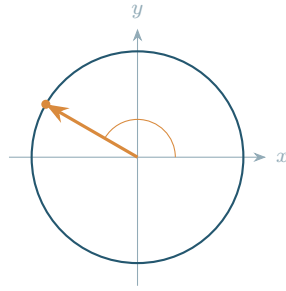
H3. Solve $\sin \theta = \frac{1}{2}$ for $0^\circ \leq \theta \leq 360^\circ$. Give all solutions.



H4. Solve $\tan \theta = 1$ for $0^\circ \leq \theta \leq 360^\circ$. Give all solutions.



H5. Solve $\cos \theta = -\frac{\sqrt{3}}{2}$ for $0^\circ \leq \theta \leq 360^\circ$. Give all solutions.



End of Chapter 6 hurdle booklet — check your working against the worked examples and ask in class if any hurdle question is unclear.